

rent limits, and protects against short circuits and over-voltage. Operating at -40°C to +105°C, it achieves up to 89% efficiency and is encapsulated in thermally conductive resin for optimal power dissipation.

*Gaia Converter*

<https://www.gaia-converter.com>

## GaN power ICs for data centres, inverters

Cambridge GaN Devices (CGD) has introduced two new packages for its



ICEGaN family of GaN power ICs, improving thermal performance and inspection simplicity.

The DHDFN-9-1 (dual heat-spreader DFN) features a 10 × 10mm footprint, dual-side cooling, and wettable flanks for optical inspection, supporting applications up to 6kW. The BHDFN-9-1 (bottom heat-spreader DFN) focuses on bottom-side cooling with a thermal resistance of 0.28 K/W, maintaining a similar footprint to TOLL packages. Enhanced thermal resistance allows higher power output, cooler operation, reduced heat sinking, and extended device lifetimes, offering cost-effective solutions for various applications.

*Cambridge GaN Devices*

<https://camgandevices.com>

## Chiplet for smart rings to monitor health and activity

Ambiq has partnered with Bravechip to develop the BCL603S3L chiplet,



incorporating the Apollo3 Blue system-on-a-chip (SoC). This chiplet

enhances smart ring designs, improving energy efficiency, functionality, performance, and reliability. Manufacturers can integrate it into smart rings for health monitoring and activity tracking, including heart rate, blood oxygen level, and step counting. The Apollo3 Blue optimises the CPU to 6µA/MHz, minimising power

usage. With BLE 5.2 and low-power Bluetooth stacks, the SoC supports real-time data via smartphones. The compact 4 × 6.8mm LGA package allows a two-layer FPC board, reducing the bill of materials by 30%.

*Ambiq*

<https://ambiq.com>

## EMBEDDED

### High-performance AI-capable IoT development kit

Qualcomm has launched the Qualcomm RB3 Gen 2 development kit for high-performance, AI-capable



IoT devices. This kit supports various IoT applications, including industrial handhelds, tablets, human-machine interface

systems, point-of-sale systems, drones, kiosks, edge AI boxes, and connected cameras. Available in vision kit (camera support up to 192MP) and core kit (main board and accessories), it is powered by the Qualcomm QCS6490 SoC. Key features include an octa-core Qualcomm Kryo 670 CPU, Adreno 643 GPU, and a hexagon processor supporting 12 TOPS, offering a 10x increase in AI processing. Connectivity options include Wi-Fi 6E and Bluetooth 5.2. The kit supports Qualcomm Linux, enabling developers to integrate pre-optimised AI models and benefit from on-device machine learning, edge computing, and a unified Linux distribution for multiple SoCs. Pre-orders are available, supported by the AI hub for enhanced on-device AI performance.

*Qualcomm*

<https://www.qualcomm.com>

### Dual-channel plug-and-play gate drivers

Power Integrations has introduced the SCALE-iFlex XLT family of dual-channel plug-and-play gate drivers for managing semiconductor modules

with up to 2300V blocking voltage. Designed for wind, energy storage, and solar installations, these gate drivers



enhance system performance and reliability through active thermal management. Key features include negative

temperature coefficient (NTC) data reporting for precise thermal management and simplified hardware design by eliminating complex isolation barriers. Comprehensive safety measures meet IEC 61800-5-1 standards, ensuring RoHS compliance. The gate drivers offer 1W output power per channel, operate at altitudes up to 4000m, and feature a robust temperature range of -40°C to 85°C.

*Power Integrations*

<https://www.power.com>

### Street-smart solar light drivers

Macon has introduced two solar street light drivers: McUD85 for 12-25W LEDs with a 12V/125 Wp solar panel, and



McUD173 for 30-50W LEDs with a 12V/250Wp solar panel. These drivers feature customisable settings for load wattages, timing, dimming, and

battery types, adjustable via selection pins. They support PIR motion sensors and include a BMS wake function.

Both models have a data output port for remote monitoring. The McUD173 also has a secondary load output for devices like CCTV cameras. Certified to IEC 62093 standards, these drivers cater to street light manufacturers, municipal governments, and commercial developers, enhancing energy efficiency and sustainability.

*Macon*

<https://macon.solar>

### SoM for network and cloud deployments

iWave introduces the iW-RainboW-G63M system on module (SoM) based